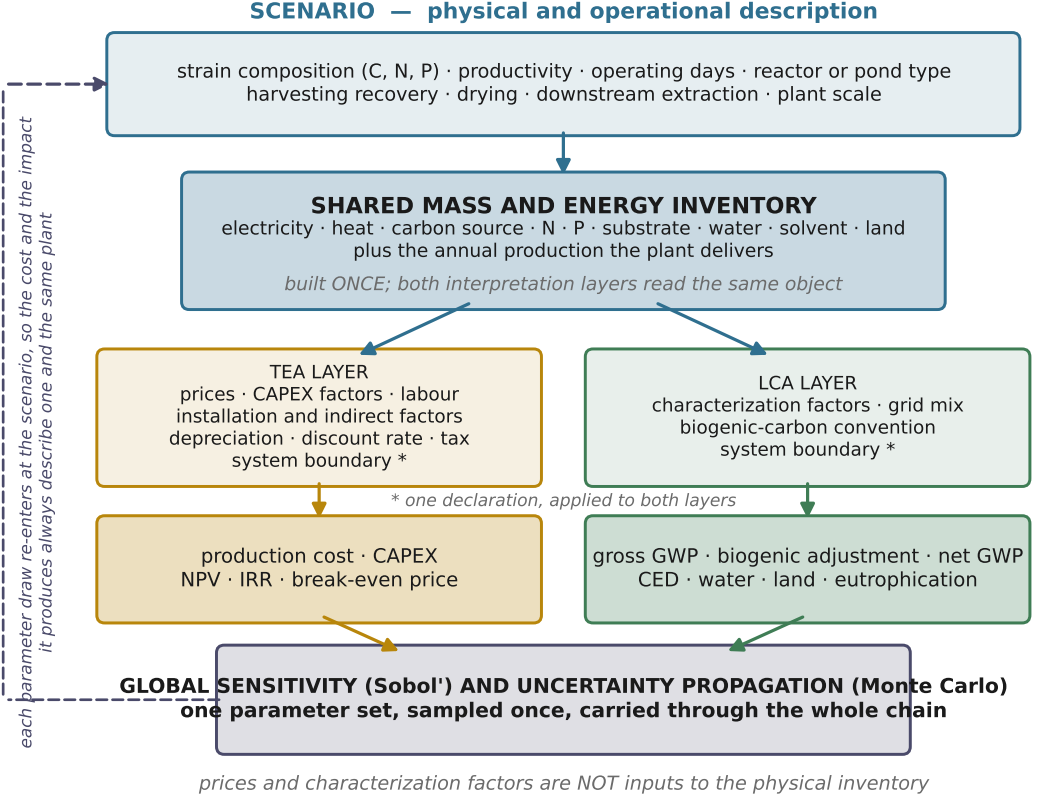
SCENARIO — physical and operational description
strain composition (C, N, P) · productivity · operating days · reactor or pond type
harvesting recovery · drying · downstream extraction · plant scale
SHARED MASS AND ENERGY INVENTORY
electricity · heat · carbon source · N · P · substrate · water · solvent · land
plus the annual production the plant delivers
built ONCE; both interpretation layers read the same object
TEA LAYER
prices · CAPEX factors · labour
installation and indirect factors
depreciation · discount rate · tax
system boundary *
LCA LAYER
characterization factors · grid mix
biogenic-carbon convention
system boundary *
* one declaration, applied to both layers
production cost · CAPEX
NPV · IRR · break-even price
gross GWP · biogenic adjustment · net GWP
CED · water · land · eutrophication
GLOBAL SENSITIVITY (Sobol') AND UNCERTAINTY PROPAGATION (Monte Carlo)
one parameter set, sampled once, carried through the whole chain
each parameter draw re-enters at the scenario, so the cost and the impact
it produces always describe one and the same plant
prices and characterization factors are NOT inputs to the physical inventory